

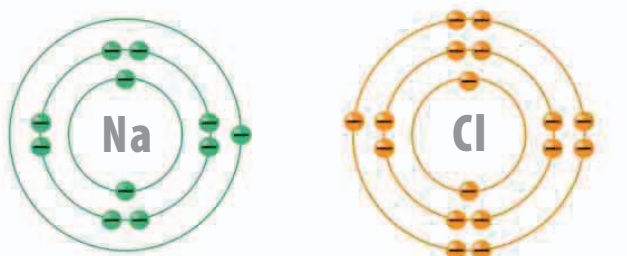
Ionic bonding

IONIC BONDING IS WHEN DIFFERENT ATOMS FORM BONDS BY GAINING OR LOSING ELECTRONS.

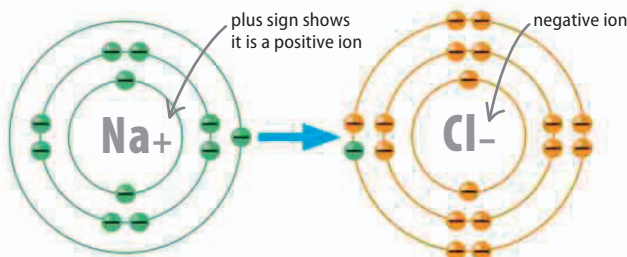
Atoms bond with each other so that they can fill the spaces in their outer electron shells. This makes them more stable.

What is an ion?

An atom has an equal number of protons and electrons so it has no overall charge. If the atom loses or gains an electron, it becomes a charged particle called an ion. Losing one electron produces a positive ion with a charge of $1+$; losing two results in a charge of $2+$. Ions formed by gaining electrons have negative charges, so gaining one electron results in a charge of $1-$.

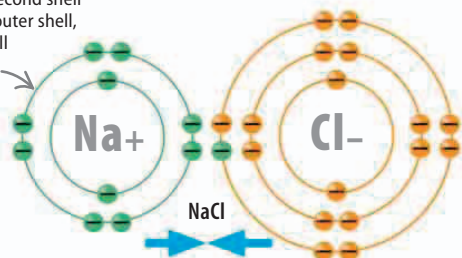


△ A sodium atom has one outer electron, while chlorine has seven electrons in the outer shell, with room for one more.



△ Sodium loses its outer electron, passing it to chlorine. Both ions now have full outer shells.

sodium's second shell is now its outer shell, which is full and stable



△ The positive charge of the sodium ion attracts the equal but negative charge of the chloride ion to form an ionic bond between the two. The resulting compound is sodium chloride (NaCl).

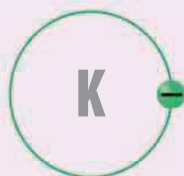
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◀ 110–111	Compounds and molecules
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Ionization energy	119 ▶
Redox reactions	132–133 ▶
Electrochemistry	148–149 ▶
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Octet rule

Atoms with low atomic numbers become full and stable when their outer shells contain eight electrons—the so-called octet rule (see page 109). Cations, or positive ions, are formed when atoms lose electrons, while anions, or negative ions, are formed when atoms gain electrons.

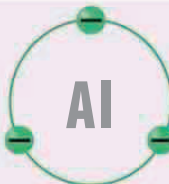
Cations (positive ions)



a potassium atom must lose one electron to form a K^+ cation

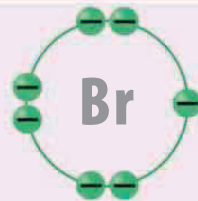


a calcium atom must lose two electrons to form a Ca^{2+} cation

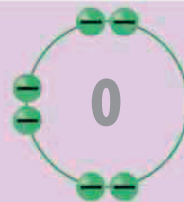


an aluminum atom must lose three electrons to form an Al^{3+} cation

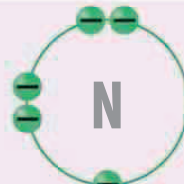
Anions (negative ions)



a bromine atom must gain one electron to form a Br^- anion



an oxygen atom must gain two electrons to form an O^{2-} anion



a nitrogen atom must gain three electrons to form a N^{3-} anion

△ Outer electrons

Atoms with between one and three outer electrons will lose them, whereas atoms with five to seven electrons will gain more. An atom with a complete outer shell is stable, so does not lose or gain electrons.